Chemistry - Graduate Level Assessment 15 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. If$_1$.0 g of rubidium and 1.0 g of bromine are reacted, what will be left in measurable amounts (more than 0.10 mg) in the reaction vessel?

   (a) RbBr only

   (b) RbBr, Rb, Br_2, and Rb_2Br

   (c) RbBr and Rb_2Br only

   (d) RbBr, Rb, and Br_2

   (e) Rb and Br_2 only

2. Cerium(III) sulfate, Ce_2(SO 4)2, is less soluble in hot water than it is in cold. Which of the following conclusions may be related to this?

   (a) The hydration energies of cerium ions and sulfate ions are very high.

   (b) The heat of solution of cerium(III) sulfate is isothermic.

   (c) The solubility of cerium(III) sulfate increases with temperature.

   (d) The heat of solution of cerium(III) sulfate is exothermic.

   (e) The hydration energies of cerium ions and sulfate ions are very low.

3. Calculate the ratio of the electrical and gravitational forces between a proton and an electron.

   (a) $4\ 10^{39}$

   (b) $6\ 10^{36}$

   (c) $7\ 10^{37}$

   (d) $5\ 10^{39}$

   (e) $9\ 10^{41}$

4. When 10.0 g of n-propylchloride is allowed to react with excess sodium in the Wurtz reaction, how many grams of hexane would be produced assuming a 70% yield?

   (a) 8.67 g

   (b) 3.82 g

   (c) 5.11 g

   (d) 3.00 g

   (e) 4.45 g

5. What is the molar mass of a monoprotic weak acid that requires 26.3 mL of 0.122 M KOH to neutralize 0.682 gram of the acid?

(a) 26.3 g mol-1

(b) 500 g mol-1

(c) 212 g mol-1

(d) 0.122 g mol-1

(e) 1.22 g mol-1

True / False (10 marks)

Answer True or False

6. True or False: The van der Waals parameter a is expressed as $8.61 \times 10^{\{} - 3\}kg\mathrm{m}^5$ s$^{\{} - 2\}mol^{\{} - 2\}inSIbaseunits.$

6. True or False: One liter of paint will cover an area of 12 square meters when brushed out to a uniform thickness of 100 microns.

7. True or False: The mass of water vapor in a 400 m^3 room at $27^and60\%relativehumidityis8.4kg.$

7. True or False: The hydrides of group-14 elements show increasing thermal stability in the order: CH 4 < SiH 4 < GeH 4 < SnH 4 < PbH 4.

8. True or False: Sulfur dioxide (SO 2) is classified as an acid anhydride due to its ability to form sulfurous acid upon reaction with water.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the frequency of a light wave that corresponds to the wavelength equal to the Bohr radius $(0.5292 \times 10^-8cm).Discussitssignificanceinquantummechanics.$

11. Discuss the concept of Hermitian operators in quantum mechanics and analyze which of the following differential operators are Hermitian, providing a rationale for your conclusions.

End of Paper